

BOON MING SHEN

Senior Embedded Software Engineer | Embedded Linux | Zephyr RTOS | Camera & ISP Systems
Singapore | +65 85019388 | boonmingshen@gmail.com | <https://boonmingshen.github.io/profile/>

PROFESSIONAL SUMMARY

Senior Embedded Software Engineer with expertise in embedded Linux, Zephyr RTOS, low-level driver development, automotive camera systems, and high-precision networking. Owns work across Yocto Linux customization, kernel tuning, V4L2 frameworks, ISP and dewarping pipelines, framebuffer output, and multi-OS IPC. Bridges hardware bring-up, system software, and AI-assisted rapid prototyping to ship robust embedded and web-based tools.

TECHNICAL SKILLS

Camera & ISP Architecture: ISP pipelines; lens dewarping and distortion correction; V4L2 framework; framebuffer architecture; video streaming; camera system integration

Low-Level & Kernel Development: Linux kernel programming; custom character/block device drivers; Yocto Project; BitBake; U-Boot; Linux device trees; firmware integration

Networking & Automotive Protocols: Precision Time Protocol (PTP); network driver architecture; Ethernet; CAN; SOME/IP; secure boot; inter-partition communication

AI & Infrastructure: Local LLM hosting (Ollama/vLLM); AI-assisted rapid prototyping; prompt engineering for software acceleration

Programming: C; C++; Python;

Validation & Tools: performance profiling; kernel tuning; memory optimization; boot-time reduction; Git; GDB; dmesg; system-analyzer; MATLAB;

WORK EXPERIENCE

MaxLinear | Singapore

Dec 2025 - Present

Senior Embedded Software Engineer

- Lead development and integration of the Precision Time Protocol (PTP) module in Zephyr RTOS for high-precision clock synchronization.
- Design, implement, and debug PTP networking features and stack components within Zephyr, ensuring robust operation on supported hardware platforms.
- Pre-Silicon Chip Testing & Verification:** Pre-silicon chip testing and validation for the Synopsys XGMAC IP, specifically validating IEEE 1588 Precision Time Protocol (PTP) hardware timestamping and network driver performance prior to tape-out.
- AI Cost-Optimization Infrastructure:** Self-hosted and configured an on-premise local LLM instance for the engineering organization, mitigating API token fees, enhancing data privacy, and accelerating internal development workflows.

Continental Automotive | Singapore

Feb 2023 - Nov 2025

Embedded Software Developer

- Developed and optimized embedded Linux software for HPC cockpit systems.
- Customized Yocto-based Linux distributions and tuned kernel and boot configuration to reduce startup time.
- Brought up camera modules and implemented Linux device drivers for camera interfaces and framebuffer-based display outputs.
- Integrated and validated Surround View Monitoring (SVM), Rear View Camera (RVC), and Driver Monitoring System (DMS) features.
- Designed low-level drivers and system software for ISP pipelines, including real-time lens distortion correction and dewarping for fisheye camera modules.
- Built camera features across V4L2 user-space frameworks, Linux kernel drivers, and low-level hardware control for RVC and SVM systems.
- Engineered real-time dynamic parking guidelines by synchronizing steering-angle telemetry from CAN bus with the ISP pipeline and injecting trajectory graphics into framebuffer output.
- Designed Inter-Partition Communication (IPC) between Automotive Controller, Linux, and Android environments.

Celestica | Malaysia

Nov 2021 - Feb 2022

Manufacturing Process Engineer Intern

- Assisted with Bill of Materials (BOM) tracking and material availability for manufacturing production processes.
- Coordinated with manufacturing and shopfloor engineering teams to troubleshoot material issues affecting production timelines.
- Supported inventory management and contributed improvements to material handling and tracking workflows.

AI-ASSISTED DEVELOPMENT & INDEPENDENT PROJECTS

Personal web profile | Live Project | Developer | [LINK](#)

- Built and deployed a responsive personal profile website to showcase professional background, technical skills, projects and contact information.

Room Rental Checklist Web Application | Live Project | Developer | [LINK](#)

- Built a lightweight client-side productivity tool to streamline housing and rental inspection workflows.
- Used AI-assisted development to implement UI features, manage state, and deploy directly to GitHub Pages.

Chess Challenge Game | Live Project | Developer | [LINK](#)

- Designed and built an interactive browser-based chess application using HTML, CSS, and JavaScript.
- Focused on clean UI design, game-state handling, rule logic, and smooth web execution.

Bit Converter Web Application | Live Project | Developer | [LINK](#)

- Designed and deployed a specialized developer tool for real-time bit manipulation, binary-hexadecimal-decimal conversions, and visual bit field inspection.

EDUCATION

Monash University: Bachelor of Electrical and Computer Systems Engineering (Hons), First Class Honours | Oct 2018 - Jun 2022

Sunway College Johor Bahru: A-Level | Jan 2017 - Jul 2018 | Math Innovation Competition, Gold Medal

LANGUAGES

English, Chinese, Malay